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Data Engineering and AI - ENGE707

Final Report for Project Phase 2

Machine Learning and LCS-Based Diabetes Risk Prediction

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Introduction

Type 2 diabetes often develops gradually and may remain undiagnosed until complications appear, so there is value in tools that can flag people who would benefit from clinical testing. Survey data describing health, lifestyle and demographic characteristics is inexpensive to collect, which makes it a practical basis for preliminary screening, although it contains no clinical measurements.

This report continues the diabetes risk prediction project developed during Phase I. Phase II focuses on preparing the data for machine learning, implementing and improving an eLCS-based classification system, and comparing its performance with conventional machine learning models. The results are evaluated using measures appropriate for the imbalanced target classes, while the learned eLCS rules are examined to understand how the system produces its predictions.

A Learning Classifier System was selected because it represents what it has learned as a population of readable IF-THEN rules. In a health screening context, being able to examine why a particular person was flagged matters as well as how often the system is correct.

The report follows the structure of the assessment tasks. Task 1 summarises Phase I. Tasks 2 to 4 establish the original eLCS baseline, describe the preprocessing applied to the data, and present the improved system. Tasks 5 and 6 set out the experimental design and compare twelve models. Tasks 7 and 8 interpret the learned rules and discuss the limitations of the work.

Task 1: Project Phase I Summary

1.1 Problem, Target Variable, and Prediction Objective

This project investigates whether self-reported health, lifestyle, and demographic information can help identify people who may have an increased risk of prediabetes or diabetes. The objective is to develop a low-cost preliminary screening system that could indicate when further clinical testing may be appropriate (World Health Organization, 2016). The system is intended to support early screening rather than diagnose diabetes late.

The project is a binary classification problem. The target variable, `Diabetes_binary`, uses 0 for respondents reporting no diabetes and 1 for respondents classified as having prediabetes or diabetes. The prediction objective is to use the available survey information to predict which of these two classes a respondent belongs to.

1.2 Dataset Overview and Characteristics

The project uses the CDC Diabetes Health Indicators dataset (Centres for Disease Control and Prevention, 2015; UCI Machine Learning Repository, 2023), which was derived from the 2015 Behavioural Risk Factor Surveillance System survey. The dataset contains 253,680 respondent records and 21 predictor variables describing health, lifestyle, and demographic characteristics.

The predictors include blood pressure and cholesterol status, BMI, general health, physical and mental health, physical activity, age, education, and income. Most variables are binary or ordered categories, while BMI is treated as a continuous numerical measurement. Age, education, income, and general health are represented by ordered category codes rather than exact measurements.

Table 1: Characteristics of the CDC Diabetes Health Indicators dataset

Attribute	Detail
Name	CDC Diabetes Health Indicators
Source	UCI Machine Learning Repository, Dataset ID 891
Link	https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators
Origin	CDC's 2015 Behavioural Risk Factor Surveillance System (BRFSS)
Size	253,680 records, 21 features + 1 target variable.
Target variable	<code>Diabetes_binary</code> (0 = no diabetes, 1 = prediabetes/diabetes)
Access date	06/08/2026
Licence	CC0: Public Domain, no usage restrictions

Phase I exploratory analysis found that diabetes prevalence was higher among several groups, including older respondents and those reporting higher BMI, high blood pressure, high cholesterol, poorer general health, and difficulty walking. These relationships were observational and did not establish that any individual factor directly caused diabetes.

1.3 Phase I Data-Quality Findings

Phase I involved inspecting, cleaning, transforming, and validating the dataset. No missing values or invalid category codes were identified. However, several characteristics required careful consideration during data preparation. A total of 24,206 rows contained the same combination of values as another row. These rows were retained because the dataset does not provide unique participant identifiers. Therefore, matching rows may represent different respondents who supplied the same answers rather than accidental duplicate records.

BMI values ranged from 12 to 98. Although these included unusually low and high observations, the values were retained because they remained possible, and there was not enough evidence to classify them as data entry errors. Age, education, income, and general health were stored as integer codes despite representing ordered categories. They were converted into ordered categorical variables during Phase I so that their meanings would be represented more accurately. This information will need to be considered again when preparing the data for machine learning models.

The target variable was unevenly distributed. Approximately 86.1% of respondents were in the no-diabetes class, while only 13.9% were in the prediabetes/diabetes class. Because of this imbalance, a model could achieve high accuracy simply by predicting "no diabetes" for most respondents, while still failing to identify many people in the smaller positive class (He & Garcia, 2009). Therefore, Phase II will use additional evaluation measures to assess how well the models perform across both classes.

1.4 Suitability for an LCS-Based Solution

This problem is suitable for a Learning Classifier System because it is a supervised binary classification problem involving a mixture of binary, ordinal, and continuous predictors (Urbanowicz & Moore, 2009). An LCS can learn condition action rules that identify combinations of characteristics associated with each target class.

For example, an LCS rule could combine conditions involving BMI, age, general health, and blood-pressure status before predicting whether a respondent belongs to the no-diabetes or prediabetes/diabetes class. The model is therefore able to represent interactions between multiple characteristics rather than considering each predictor separately.

The rule-based structure of an LCS may also make its predictions easier to examine and explain than those of some less transparent models. This is useful in a health-related screening context, where understanding the conditions contributing to a prediction is important. Phase II will therefore evaluate an original eLCS model, develop an improved LCS-based system, and compare its performance with conventional machine learning models.

Task 2: Original LCS System on raw Dataset

2.1 Selection of the Baseline LCS

The original eLCS implementation from the scikit-eLCS package (Urbanowicz, 2020) was selected as the baseline Learning Classifier System. eLCS was appropriate because the task is a binary classification problem, and the resulting rule population can later support interpretation of the model's decisions (Urbanowicz & Browne, 2017).

The original eLCS algorithm was used without modification. A fixed random seed of 42 was added to make the experiment reproducible. This baseline provides a reference against which the improved system can later be compared.

2.2 Dataset and Minimal Compatibility Processing

The Phase I cleaned diabetes dataset was used as the minimally processed input. It contained 253,680 records, 21 predictor variables, and the binary target variable Diabetes_binary. The target distinguishes respondents with no diabetes from respondents with prediabetes or diabetes.

An 80/20 stratified training and test split was applied using a random seed of 42. This produced 202,944 training records and 50,736 test records while preserving the original class distribution of approximately 86.07% no diabetes and 13.93% prediabetes/diabetes.

Only the processing required for compatibility with eLCS was performed. The predictor and target data were converted from pandas objects into NumPy arrays before model training. No scaling, feature engineering, class balancing, or resampling was applied because the purpose was to measure the performance of the original system before improvements.

2.3 Baseline eLCS Configuration

The baseline used the original eLCS parameter settings. The main parameters were:

Table 2: Default eLCS parameter settings used for the baseline

Parameter	Value
Learning iterations	10000
Population size (N)	1000
Initial specified-attribute probability (p_spec)	0.5
Discrete attribute limit	10
Fitness exponent (nu)	5
Crossover probability (chi)	0.8
Mutation probability (mu)	0.04
GA threshold (theta_GA)	25
Deletion threshold (theta_del)	20
Subsumption threshold (theta_sub)	20

Subsumption accuracy (acc_sub)	0.99
Learning rate (beta)	0.2
Selection method	Tournament
Tournament proportion (theta_sel)	0.5
GA subsumption	Enabled
Correct-set subsumption	Disabled
Random seed	42

The model required 7.87 seconds for training. Generating predictions for all 50,736 test records required 39.52 seconds, and no records were left unclassified.

2.4 Baseline Performance

The baseline results were:

Table 3: Baseline eLCS performance on the held-out test set

Metric	Result
Accuracy	86.06%
Balanced accuracy	50.00%
Precision for prediabetes/diabetes	0.00%
Recall for prediabetes/diabetes	0.00%
F1-score for prediabetes/diabetes	0.00%

The confusion matrix was:

Table 4: Confusion matrix for the baseline eLCS on the test set

	Predicted no diabetes	Predicted prediabetes/diabetes
Actual no diabetes	43,664	3
Actual prediabetes/diabetes	7,069	0

The model predicted 50,733 respondents as having no diabetes and only three as having prediabetes or diabetes. All three positive predictions were incorrect. Therefore, the model failed to correctly identify any of the 7,069 positive cases in the test set.

2.5 Interpretation of the Baseline

Although the baseline achieved an accuracy of 86.06%, this result is misleading because the dataset is strongly imbalanced. A model that mostly predicts the majority no diabetes class can achieve high overall accuracy without successfully identifying the minority class.

The balanced accuracy of 50.00%, together with zero precision, recall and F1-score for the positive class, shows that the original eLCS configuration did not learn an effective distinction between the two target classes. This establishes a clear baseline and justifies the class-imbalance handling and preprocessing introduced in Task 3. The improved system should be evaluated mainly through balanced accuracy, recall, and F1-score rather than accuracy alone.

Task 3: Data Preprocessing and Feature Engineering

3.1 Data Quality Validation

The cleaned Phase I dataset was checked again before preprocessing to ensure that it was suitable for model development. It contained 253,680 records and 22 columns, including 21 predictors and the binary target variable. No missing values or invalid category codes were identified, and all variables had the expected data types and value ranges.

A total of 24,206 rows contained identical responses. These were retained because the dataset does not include a participant identifier, meaning they could represent different respondents who provided the same answers. The BMI values ranged from 12 to 98. Although the IQR method (Tukey, 1977) identified 9,847 potential outliers, these values were also retained because there was no evidence that they were data entry errors.

3.2 Training and Test Split

The dataset was divided into training and test sets using the same configuration as the original eLCS baseline. An 80/20 stratified split with a fixed random seed of 42 produced 202,944 training records and 50,736 test records. Stratification preserved the original class distribution of approximately 86.07% no diabetes and 13.93% prediabetes/diabetes in both sets.

The same split was also used to support a fair comparison between the original and improved systems. The test set was kept unchanged throughout preprocessing. Any operation that learned from or altered the class distribution was applied only to the training data, preventing information from the test set from influencing model development.

3.3 Class Imbalance Handling

The training set was strongly imbalanced, containing 174,667 no diabetes records and 28,277 prediabetes/diabetes records. This imbalance contributed to the original eLCS model predicting almost every test record as belonging to the no diabetes class.

Random under sampling was therefore applied to the majority class using a fixed random seed of 42 (He & Garcia, 2009). This produced a balanced training set containing 28,277 records from each class, or 56,554 records in total. Only the training data was balanced, whereas the test set retained its original class distribution. This approach gives eLCS equal exposure to both outcomes and should improve its ability to learn rules representing the minority class, although removing majority class records may result in some loss of information.

3.4 Synthetic Oversampling as an Alternative Strategy

Because random undersampling discards majority class records, SMOTENC was prepared as an alternative (Chawla et al., 2002; Lemaître et al., 2017). Ordinary SMOTE was not used because continuous interpolation could create invalid values for the dataset's discrete

variables, such as fractional smoking, age category or unhealthy day values, reducing the validity and interpretability of eLCS rules.

SMOTENC supports mixed feature types. All binary, ordinal and count based predictors were declared categorical to preserve permitted values, while BMI was treated as continuous. It was applied only to the training data and increased the minority class from 28,277 to 17,4667 records, producing 349,334 balanced training records. Validation found no missing values or invalid categories, and BMI remained within its observed range.

The smaller undersampled dataset was used for the improved eLCS because of its lower computational cost. SMOTENC was retained for the Task 6 comparison, where random undersampling achieved the highest balanced accuracy across the three conventional model types.

3.5 Feature Representation

All 21 predictors were retained because they contained valid values and could provide useful information for rule discovery. The 14 binary predictors kept their existing 0/1 encoding, while the ordered codes for general health, age, education, and income were preserved. BMI, mental health days, and physical health days also remained in their original numerical form.

No additional scaling was applied because eLCS can form range-based conditions for numerical attributes and does not rely on distance calculations between records. Preserving the original representation also keeps the resulting rules easier to interpret. No additional features were created at this stage, avoiding unnecessary complexity and allowing the effect of training class balancing to be evaluated more clearly.

3.6 Prepared Data and Expected Effect on eLCS

The preprocessing pipeline produced three final datasets: a balanced training dataset containing 56,554 records, a SMOTENC oversampled training dataset containing 349,334 records, and an unchanged test dataset containing 50,736 records. All three datasets contain the same 21 predictors and the Diabetes_binary target variable. These files provide reproducible inputs for developing and evaluating the improved eLCS system.

Balancing the training data is expected to help eLCS discover more rules for the prediabetes/diabetes class. This should improve minority class recall, F1-score, and balanced accuracy when compared with the original baseline. However, it may also reduce ordinary accuracy or increase the number of false positive predictions. The actual effect will therefore be determined by evaluating the improved model on the unchanged test set in the following tasks.

Table 5: Data-quality issues, preprocessing decisions, and justifications

Issue	Decision	Justification
Missing or invalid values	No correction required	No missing values or invalid codes were found

Identical rows	Retained	No participant identifier exists to confirm duplicates
BMI outliers	Retained	Values were unusual but not confirmed as errors
Class imbalance	Randomly under-sampled training majority class	Gives eLCS equal exposure to both classes
Feature encoding	Existing numerical representations retained	Values were already valid and suitable for eLCS
Data leakage	Test set kept unchanged	Supports fair evaluation against the baseline
Choice of oversampling method	SMOTENC rather than SMOTE	Nearly all predictors are binary or ordinal codes; SMOTE interpolation would generate impossible values and distort the eLCS rules

Task 4: Improved LCS-Based System

4.1 Improvements to the Original System

The original eLCS algorithm was kept, but the training pipeline and configuration were altered; three improvements were introduced.

First, the balanced training dataset that had been prepared in Task 3 was used; it included 56,554 records, 28,770 of which belonged to each target class. The held-out test set was left unchanged at 50,736 records, just as the original class distribution had been.

Second, the predictor types were clearly set up. The 14 binary predictors were treated as discrete attributes, while BMI, general health, mental health days, physical health days, age, education, and income were treated as range-based attributes. This enabled eLCS to create interval-base rules for numerical and order variables rather than treating their values as unrelated categories.

Third, the population size and the probability of attribute specification in eLCS were adjusted using a separate validation set (Urbanowicz & Browne, 2017). The original eLCS source code was left unaltered, and all the improvements were achieved through preprocessing and by making use of the parameters already supported by the existing implementation.

4.2 Validation and the choice of hypermeters

The dataset, which was balanced, was split into a 80% tuning training set and a 20% validation set by means of stratified sampling with a random seed set at 42. This resulted in 45,243 records in the tuning-training set and 11,311 in the validation set. Both sets maintained an equal distribution between the two target classes.

The following four configurations were assessed, with balanced accuracy serving as the criterion for selection since it treats both classes equally.

Table 6: eLCS configurations evaluated on the validation se

configuration	Population size	P_spec	Validation balance accuracy	Recall	F1-score
1	1000	0.10	0.7285	0.7436	0.7325
2	1000	0.25	0.7258	0.7880	0.7419
3	2000	0.10	0.7333	0.7795	0.7450
4	2000	0.25	0.7296	0.7843	0.7436

Configuration 3 obtained the highest balanced accuracy in the validation stage and was therefore chosen as the final model. The fact that its population size was 2,000 enabled eLCS to keep a bigger and more diverse set of rules. The lower value of p_spec, which was 0.10, promoted simpler rules that included fewer specified attributes and thus reduced the chance of there being overly specific rules.

4.3 Improved-System Procedure

The improved system followed this procedure:

1. Load the balanced training dataset and the test dataset without making any changes.
2. Remove the target variable from the 21 predictors.
3. Split the balanced training data into tuning-training and validation sets.
4. Set the binary variables as discrete and the seven numerical or ordered variables as range-based attributes.
5. Tune the candidate for eLCS configurations using the tuning-training set.
6. Assess each configuration using the validation set by means of balanced accuracy.
7. Choose the configuration that has the highest balanced accuracy based on the validation results.
8. Use the full balanced training dataset to retrain the chosen configuration.
9. Once evaluate the final model with the test set that has not been changed.
10. Compare the results with those of the original eLCS baseline.

The test set was left out when carrying out class balancing, attribute selection, and hyperparameter tuning. This ensured that information from the test set did not have any influence on the development of the model and therefore allowed for a fair final comparison.

4.4 The Final Refinement of eLCS Performance

The model employed 10,000 learning iterations, had a population size of 2,000, used an attribute specification probability of 0.10, and was given a random seed value of 42. The training process took 73.66 seconds and prediction for all 50,736 test records took 946.19 seconds. All the records were classified.

Table 7: Comparison between the original and improved eLCS

Metric	Original eLCS baseline	Improved eLCS
Accuracy	0.8606	0.7085
Balanced accuracy	0.5000	0.7309
Precision	0.0000	0.2913
Recall	0.0000	0.7621
F1-score	0.0000	0.4215

The confusion matrix that had been improved included 30,559 true negatives, 13,108 false positives, 1,682 false negatives, and 5,387 true positives.

The enhanced model succeeds in identifying 5,387 of the 7,069 individuals who were in the positive class, whereas the original baseline found no positive cases at all. Balanced accuracy rose from 0.5000 to 0.7309, and recall increased from 0.0000 to 0.7621.

4.5 The Interpretation of the Improvement

The fall in ordinary accuracy from 0.8606 to 0.7085 does not indicate that the improved model was less useful. The original model had attained a high level of accuracy by almost predicting every respondent as not having diabetes, and since its recall for the positive class was zero, it proved to be an ineffective means of screening for both classes.

The enhanced model resulted in a higher number of false positives and thus its precision fell to 0.2913. At the same time, the number of false negatives decreased greatly, with about 76% of the positive cases being detected. This kind of trade-off might be suitable for use in initial screening since a false positive cause the case to be subject to further evaluation, whereas a false negative could mean that a person who is at risk would not be identified.

The main gain was achieved by balancing the training data. With the original eLCS configuration used on the balanced and preprocessed data, a balanced accuracy of about 0.7302 was obtained; this was then very slightly increased to 0.7309 because of adjusting the attribute configuration and tuning the hyperparameters. Hence, the major improvement stemmed from giving eLCS equal exposure to both target classes, while the tuning contributed a minor further improvement.

The new system gave rise to much more balanced predictions and was considerably better at identifying the prediabetes/diabetes category than the original baseline.

Task 5: Experiments

5.1 Validation Strategy

All models were evaluated through a shared evaluation module, `src/evaluation.py`, so that every result in this report was produced by the same code on the same records. The module fixes the split, the cross-validation folds, and the metric definitions, which removes the risk of differences between models being caused by differences in how they were evaluated.

A stratified 80/20 train-test split with a fixed random seed of 42 was used, producing 202,944 training records and 50,736 test records. Stratification preserved the class distribution in both sets. This split was chosen rather than cross-validation alone for two reasons. First, it provides a completely held-out set of 50,736 records that no model saw during development, including hyperparameter tuning, which supports an unbiased final comparison. Second, eLCS is computationally expensive: one improved-eLCS run required approximately 74 seconds to train and 946 seconds to predict, so cross-validating three LCS configurations would have required around eight hours.

Stratified 10-fold cross-validation was additionally applied within the training set for the conventional models, producing ten scores per model for statistical testing. Ten folds were used rather than five because the Wilcoxon signed-rank test with five paired observations cannot produce a two-sided p-value below 0.0625 (Demšar, 2006). So, no comparison could reach significance at the 5% level regardless of the difference between models.

5.2 Selection of Evaluation Metrics

Accuracy is not a suitable primary metric for this dataset (He & Garcia, 2009). As Task 2 demonstrated, the baseline eLCS achieved 86.06% accuracy while failing to identify a single positive case, because predicting the majority class for every record reproduces the class distribution.

The following metrics were therefore recorded for every model:

Table 8: Evaluation metrics and justification for inclusion

Metric	Reason for inclusion
Balanced accuracy	Primary metric; averages recall across both classes, so majority-class prediction scores 50%
Recall	Proportion of prediabetes/diabetes cases detected; a missed case is the most costly error in screening
Precision	Proportion of positive predictions that are correct; indicates the burden of unnecessary follow-up testing
F1-score	Summarises the balance between precision and recall
Specificity	Proportion of no-diabetes cases correctly identified
ROC-AUC	Threshold-independent ranking quality
PR-AUC	Ranking quality focused on the minority class, more informative than ROC-AUC under imbalance (Saito & Rehmsmeier, 2015).

Records left unclassified by an LCS, where no rule in the population matches, were counted as negative predictions and reported separately. No records were left unclassified in any run.

5.3 Statistical Testing

Three tests were applied, addressing different questions.

Friedman test (Demšar, 2006): Applied to the ten cross-validation balanced-accuracy scores of the nine conventional model variants, to establish whether any of the models differed. The test is non-parametric, which suits fold scores that cannot be assumed normally distributed, and it accounts for the scores being paired across identical folds. The result was $\chi^2 = 78.05$, $p < 0.000001$, so the null hypothesis of equal performance was rejected. Mean ranks placed Random Forest with under sampled training data first at 1.4, and Gaussian Naive Bayes with class weighting last at 9.0.

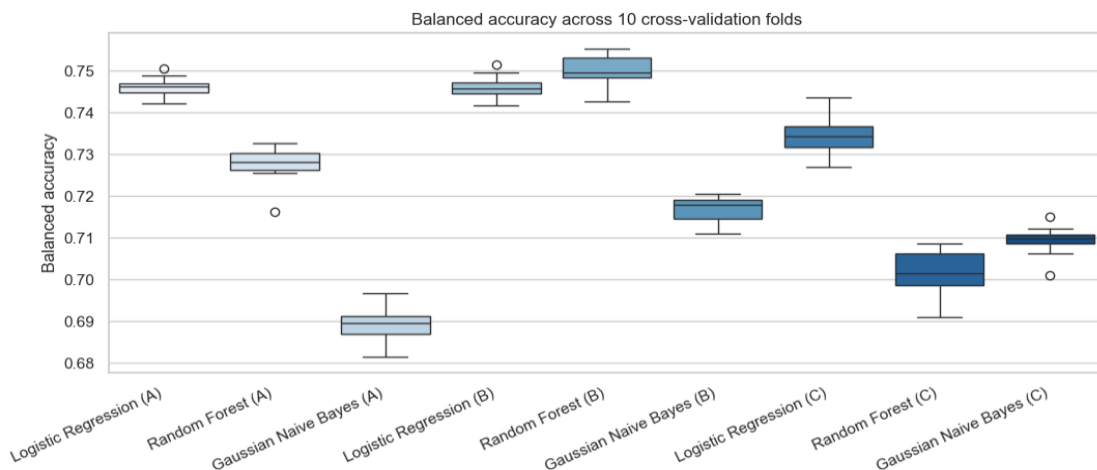


Figure 1: Balanced accuracy across 10 cross-validation folds

The figure shows why the fold scores support statistical testing: the spread within each model is narrow relative to the differences between models. Fold standard deviations ranged from 0.0024 for Logistic Regression (A) to 0.0055 for Random Forest (C), so the ranking is stable across resamples rather than an artefact of any particular partition. The separation between Random Forest (B) and the three Gaussian Naive Bayes variants is visible without testing, whereas its advantage over the two Logistic Regression variants is small enough that the statistical tests are needed to establish it.

Wilcoxon signed-rank tests against a control: Since the Friedman test identifies only that a difference exists, pairwise tests were required. All 36 pairs were tested first, with Holm-Bonferroni correction (Holm, 1979), and no comparison reached significance. This was a property of the procedure rather than of the models: with ten folds the smallest attainable two-sided p-value is 0.00195, and multiplying by 36 comparisons gives 0.070, so significance was unreachable even where one model won on all ten folds.

Each model was therefore compared against the best-performing model as a control, which is the standard post-hoc procedure in this situation (Demšar, 2006) and reduces the correction factor from 36 to 8. All eight comparisons were significant after correction:

Table 9: Wilcoxon signed-rank tests against Random Forest (B) as control, with Holm correction

Comparison against Random Forest (B)	Mean difference	Folds won	p (Holm)
Gaussian Naive Bayes (A)	0.0606	10/10	0.0156
Random Forest (C)	0.0479	10/10	0.0156
Gaussian Naive Bayes (C)	0.0403	10/10	0.0156
Gaussian Naive Bayes (B)	0.0327	10/10	0.0156
Random Forest (A)	0.0221	10/10	0.0156
Logistic Regression (C)	0.0147	10/10	0.0156
Logistic Regression (B)	0.0035	8/10	0.0273
Logistic Regression (A)	0.0035	8/10	0.0273

Both all-pairs and control-based results were retained in the analysis notebook, as the contrast between them illustrates the effect of the correction procedure on the conclusions that can be drawn.

McNemar's test (Dietterich, 1998): Applied to the held-out test predictions of all twelve models, comparing them record by record rather than through aggregate scores. This test was necessary because the LCS models were not cross-validated and therefore have no fold scores, but their test-set predictions were saved and could be compared directly. With 50,736 paired observations the test has considerably more statistical power than a ten-fold Wilcoxon test. Holm-Bonferroni correction was applied across all 66 pairs.

5.4 Interpretation of Statistical Significance

Sixty-one of the 66 McNemar comparisons were significant after correction. The five that were not are informative:

Table 10: McNemar comparisons that were not significant after Holm correction

Comparison	p (Holm)	Interpretation
Improved eLCS vs original eLCS on preprocessed data	0.0511	The improvement from tuning is not statistically distinguishable
Logistic Regression A vs B vs C (three pairs)	1.0000	The imbalance strategy has no effect on a linear model
Gaussian Naive Bayes (A) vs Random Forest (C)	1.0000	Equal error rates despite different errors

The first result is the most consequential for this project and is discussed in Tasks 6 and 8. The Logistic Regression results are consistent with theory: class weighting and resampling both alter the effective class prior, and for a linear decision boundary the two are close to equivalent, whereas for a tree ensemble the available split points change with the records present.

The final comparison shows why McNemar's test is preferable to comparing accuracies. Gaussian Naive Bayes (A) and Random Forest (C) disagreed with each other on 10,490 records, 5,250 in favour of one model and 5,240 in favour of the other. The two models are equally accurate while being wrong about substantially different people, so equal performance does not imply equivalent behaviour.

Finally, statistical significance should not be equated with practical importance. With a test set of 50,736 records, differences too small to affect a screening decision can be detected reliably. The difference between Random Forest (B) and Logistic Regression (A) was 0.0035 in balanced accuracy, which is statistically significant but corresponds to fewer than 20 additional correct classifications in the test set.

Task 6: Model Comparison

6.1 Models Included in the Comparison

Twelve models were compared. These cover the three LCS configurations required by the assessment and three conventional machine learning models (Breiman, 2001; Pedregosa et al., 2011), each run under three different class-imbalance strategies.

Table 11: The twelve models included in the comparison

Group	Models
LCS	Original eLCS on the minimally processed data, original eLCS on the preprocessed data, improved eLCS
Conventional	Logistic Regression, Random Forest, Gaussian Naive Bayes
Imbalance strategies (conventional only)	A = class weighting on the full training set, B = random undersampling, C = SMOTENC oversampling

All three conventional models are non-deep-learning, as the assessment requires. Every model was trained and tested on the same records and scored by the same code, so the results can be compared directly.

Running each conventional model under all three imbalance strategies serves a second purpose. The LCS models were trained on the undersampled data, which is much smaller than the full training set. If the conventional models had been given all 202,944 training records while the LCS received only 56,554, the comparison would have favoured them for reasons unrelated to the algorithms.

6.2 Results

Table 12 reports the held-out test performance of all twelve models, ordered by balanced accuracy. Letters A, B and C denote the class-imbalance strategy used by the conventional models: A = class weighting, B = random undersampling, C = SMOTENC.

Table 12: Held-out test performance of all twelve models, ordered by balanced accuracy. A = class weighting, B = random undersampling, C = SMOTENC

Model	Accuracy	Balanced accuracy	Precision	Recall	F1	Specificity	ROC-AUC	PR-AUC
Random Forest (B)	0.7184	0.7486	0.3038	0.7905	0.4389	0.7068	0.8229	0.4146
Logistic Regression (B)	0.7317	0.7447	0.3112	0.7628	0.4421	0.7267	0.8195	0.3924
Logistic Regression (A)	0.7316	0.7439	0.3108	0.7611	0.4414	0.7268	0.8196	0.3926
Logistic Regression (C)	0.7320	0.7350	0.3078	0.7390	0.4346	0.7309	0.8066	0.3733
Improved eLCS	0.7085	0.7309	0.2913	0.7621	0.4215	0.6998	0.8008	0.3635
Original eLCS (preprocessed)	0.7057	0.7302	0.2894	0.7640	0.4198	0.6963	0.7982	0.3578
Random Forest (A)	0.7945	0.7242	0.3626	0.6268	0.4594	0.8216	0.8214	0.4158
Gaussian Naive Bayes (B)	0.7255	0.7152	0.2955	0.7009	0.4157	0.7295	0.7805	0.3351

Gaussian Naive Bayes (C)	0.6060	0.7061	0.2402	0.8448	0.3740	0.5673	0.7672	0.3396
Random Forest (C)	0.7720	0.7015	0.3274	0.6039	0.4246	0.7992	0.7976	0.3635
Gaussian Naive Bayes (A)	0.7722	0.6849	0.3198	0.5639	0.4082	0.8059	0.7803	0.3353
Original eLCS (minimally processed)	0.8606	0.5000	0.0000	0.0000	0.0000	0.9999	0.6911	0.2495

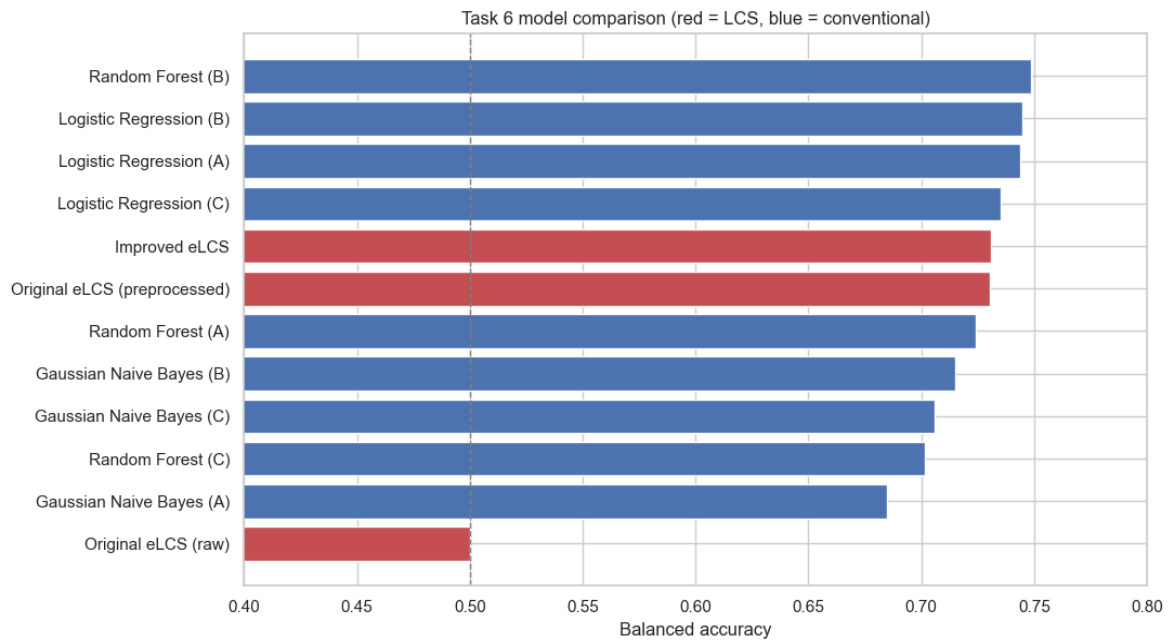


Figure 2: Balanced accuracy of all models

Figure 2 shows the balanced accuracy of all twelve models on the held-out test set. LCS models are shown in red, conventional models in blue. The dashed line at 0.50 marks the level expected from predicting a single class for every record.

6.3 The Effect of the Imbalance Strategy

Random undersampling gave the best result for all three conventional models, which was not the expected outcome, since it discards about 146,000 majority-class records.

Table 13: Balanced accuracy of each conventional model under the three imbalance strategies

Model	A: class weighting	B: undersampling	C: SMOTENC
Logistic Regression	0.7439	0.7447	0.7350
Random Forest	0.7242	0.7486	0.7015
Gaussian Naive Bayes	0.6849	0.7152	0.7061

Two points follow from this. First, training on the smaller balanced set did not disadvantage the models, so the LCS was not penalised by the preprocessing choice, and the comparison is fair. Second, SMOTENC performed worst for two of the three models despite generating far more

training data, which suggests the synthetic minority records added noise rather than useful information.

The strategy also mattered far more for Random Forest than for Logistic Regression. McNemar's test found no significant difference between the three Logistic Regression variants, while Random Forest improved by 0.0244 when moving from class weighting to undersampling. A linear model responds to class weighting in much the same way as to resampling, whereas a tree ensemble depends on which records are actually present when choosing its splits (Breiman, 2001).

6.4 Comparing the LCS Models

The three LCS configurations show where the improvement came from.

Table 14: Performance of the three eLCS configurations

Configuration	Balanced accuracy	Recall	Cases detected
Original eLCS, minimally processed data	0.5000	0.000	0 of 7,069
Original eLCS, preprocessed data	0.7302	0.764	5,401
Improved eLCS	0.7309	0.762	5,387

Balancing the training data raised balanced accuracy by 0.2302. The attribute typing and hyperparameter tuning added a further 0.0008 on top of that.

McNemar's test confirms this. The improved eLCS and the original eLCS on preprocessed data were the only pair among the twelve models that did not differ significantly ($p = 0.0511$ after correction). The two models disagreed on 2,971 records, but the disagreements were almost evenly split, 1,556 in favour of the improved model and 1,415 against.

In other words, nearly all of the improvement came from the preprocessing in Task 3 rather than from the changes to the LCS itself. This is discussed further in Task 8.

There is some evidence that the tuning improved the model in a way the threshold metrics do not capture. ROC-AUC rose from 0.7982 to 0.8008 and PR-AUC from 0.3578 to 0.3635, so the improved model ranks cases slightly better even though its final decisions are nearly the same.

The baseline result is also worth a second look. Although it detected no cases at all, its ROC-AUC was 0.6911, which is above the 0.5 expected from random guessing. The model had learned something about the data; what it failed to do was act on that information when assigning classes, because the rules covering the majority class dominated the population.

6.5 LCS Compared with the Conventional Models

The improved eLCS reached 0.7309 balanced accuracy, placing it fifth of the twelve models. It performed better than all three Gaussian Naive Bayes variants and two of the three Random

Forest variants, but worse than the best Random Forest at 0.7486 and both of the better Logistic Regression variants at about 0.744. McNemar's test found all of these differences significant.

Recall, which matters most for screening, tells a similar story. The improved eLCS detected 5,387 of 7,069 cases (76.2%), close to Logistic Regression at 76.3% but below Random Forest (B) at 79.1%.

The difference in computational cost is much larger than the difference in performance as shown in Table 15:

Table 15: Training and prediction times compared with balanced accuracy

Model	Training	Prediction	Balanced accuracy
Logistic Regression (B)	0.1 s	0.008 s	0.7447
Random Forest (B)	1.1 s	0.4 s	0.7486
Improved eLCS	12.7 s	224.7 s	0.7309

The improved eLCS took roughly 28,000 times longer to generate predictions than Logistic Regression and scored lower. On predictive performance alone, there is no case for using the LCS on this dataset. Its advantage lies in producing an explicit set of rules that can be read and checked, which is examined in Task 7.

6.6 Metrics Do Not Agree on a Single Best Model

The ranking depends on which metric is used, which supports the decision in Task 5 to report several.

Random Forest (A) achieved the highest PR-AUC of 0.4158 but ranked seventh on balanced accuracy, because it makes fewer positive predictions overall. Gaussian Naive Bayes (C) achieved the highest recall of 0.8448 while having the lowest accuracy at 0.6060, since it flagged 24,866 people to find 5,972 cases.

Precision was low for every model, ranging from 0.24 to 0.36. Even the best model, Random Forest (B), flagged 18,392 people to identify 5,588 true cases, so roughly two out of every three people it flags do not have diabetes. This is a property of the problem rather than of any particular model: the condition affects about 14% of the population, and the survey features available are only moderately predictive. The practical consequences are discussed in Task 8.

Task 7: Interpretation of Results

7.1 What the Comparison Showed

Three findings from Task 6 matter for interpreting the system.

The original eLCS on minimally processed data learned almost nothing usable. It detected none of the 7,069 positive cases, although its ROC-AUC of 0.6911 shows it had captured some signal it could not act on.

Balancing the training data changed that completely, raising balanced accuracy from 0.5000 to 0.7302. The further changes in the improved system, explicit attribute typing and hyperparameter tuning, added only 0.0008, a difference McNemar's test found non-significant ($p = 0.0511$). The preprocessing, not the modifications to the LCS, produced nearly all of the improvement.

The improved eLCS also did not match the best conventional models, reaching 0.7309 against 0.7486 for Random Forest. The remainder of this section examines what the LCS offers in return.

7.2 The Structure of the Rule Population

The improved eLCS produced 1,821 rules: 1,076 predicting diabetes or prediabetes and 745 predicting no diabetes. Each rule specifies conditions on a subset of the 21 features, predicts a class, and carries its own accuracy and match statistics.

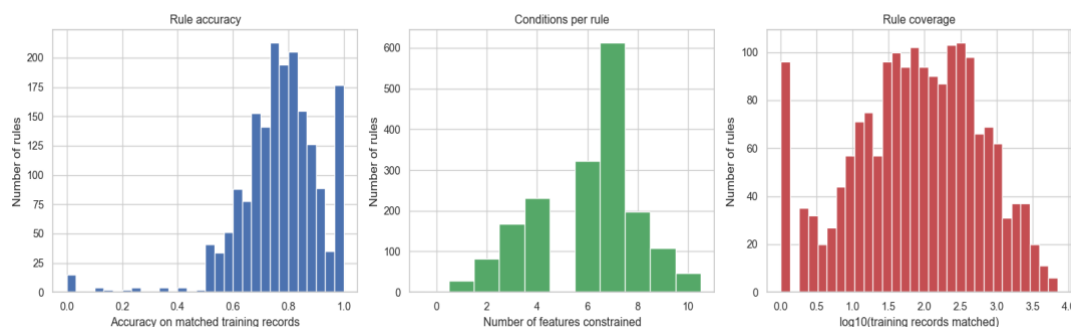


Figure 3: Distribution of accuracy, number of conditions, and coverage across the 1,821 rules in the final population.

Rules constrain between 1 and 10 of the 21 features. Those with few conditions match many records and make general statements; those with many conditions describe narrow subgroups. Coverage is uneven, with a median of 85 matched training records per rule.

7.3 Sample Rules

Rules were selected on accuracy and coverage together, since a rule matching only a few records explains very little even if it is perfectly accurate. Rules matching fewer than 200 training records were excluded, leaving 647 for consideration.

Rule 1: predicting diabetes (85.9% accurate, 811 records)

IF high blood pressure AND BMI between 30.0 and 70.0 AND not a heavy drinker AND has healthcare coverage AND general health rating at least 2.7 AND difficulty walking or climbing stairs THEN diabetes or prediabetes

This is the highest-coverage accurate rule in the population, and it describes a recognisable clinical picture: hypertension, obesity, poor self-rated health and reduced mobility. The BMI lower bound evolved to 30.0, which is the standard clinical threshold for obesity. The system recovered that cut-point from the data without being given it.

Rule 2: predicting no diabetes (93.9% accurate, 374 records)

IF no high blood pressure AND no high cholesterol AND not a smoker AND no heart disease AND eats vegetables daily AND not a heavy drinker AND general health rating at most 2.5 AND no difficulty walking AND female THEN no diabetes

This rule works by ruling out the metabolic conditions that commonly occur alongside type 2 diabetes (Alberti et al., 2009). Its direction agrees with the Phase I exploratory analysis, where diabetes prevalence was lower among respondents reporting good general health and no mobility difficulty.

Rule 3: a weak rule (51.3% accurate, 940 records)

IF not a heavy drinker THEN diabetes or prediabetes

This single-condition rule is barely better than chance, yet it was retained and matches a large number of records. Its presence is not an error. Predictions in eLCS come from the collective vote of all rules matching a record, weighted by accuracy and fitness, so weak rules contribute little without being removed. It does mean, however, that individual rules cannot be read as clinical claims in isolation.

One structural pattern is worth noting. The most accurate rules have a numerosity of 1, meaning a single copy in the population, while broader and weaker rules carry more copies. The system reinforces general rules that fire often, whereas the sharpest rules remain specific and rare.

7.4 Which Features the Rules Rely On

A rule population does not produce a single importance score in the way a tree ensemble does. An equivalent view is obtained by counting how often each feature is constrained, weighted by rule numerosity.

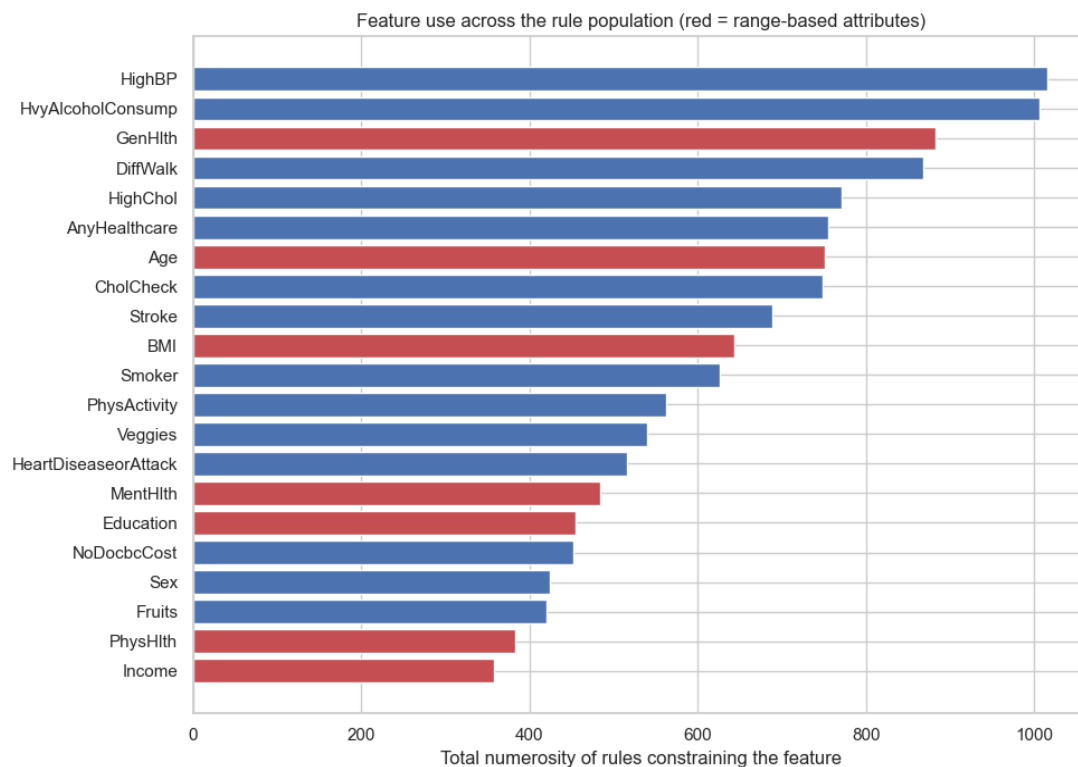


Figure 4: Feature use across the rule population, weighted by numerosity. Range-based attributes are shown in red.

Moreover, Table 16 below shows the features across the 1,821 rules:

Table 16: The seven most frequently used features across the 1,821 rules.

Feature	Rules using it	Share of population
HighBP	928	51.0%
HvyAlcoholConsump	926	50.9%
GenHlth	805	44.2%
DiffWalk	797	43.8%
HighChol	704	38.7%
AnyHealthcare	689	37.8%
Age	693	38.1%

High blood pressure leading the population is consistent with clinical understanding, as are general health, walking difficulty and high cholesterol. Age and BMI also appear frequently.

One result does not fit. Heavy alcohol consumption is the second most used feature, appearing in 926 rules, despite not being an established diabetes predictor. A plausible explanation is that the variable is rare and highly skewed, so it separates records cleanly without being causally related to the outcome. This is a caution about reading feature usage as importance: the count reflects what the search found useful for partitioning the data, not what is clinically meaningful.

7.5 Are the Explanations Trustworthy?

Interpretability is only valuable if the explanations are reliable. Three checks were applied.

Agreement with clinical knowledge: Where a rule constrains an established risk factor to be present, the great majority of such rules predict diabetes.

Table 17: Direction of prediction when a known risk factor is present in a rule, weighted by numerosity.

Risk factor	Predicts diabetes	Predicts no diabetes	Share predicting diabetes
Heart disease or attack	199	6	97.1%
High blood pressure	576	19	96.8%
Difficulty walking	331	15	95.7%
High cholesterol	350	17	95.4%
Stroke	50	5	90.9%

The population is directionally consistent with established knowledge, which supports the credibility of its explanations.

Accuracy spread: 41.7% of rules exceed 80% accuracy and 9.2% fall below 60%, with a numerosity-weighted mean of 0.772. Quality is therefore uneven, and a reader inspecting a single rule at random could encounter one of the weak ones.

Coverage concentration: 126 rules, 6.9% of the population, account for half of all matches. A useful summary of the model is therefore far smaller than 1,821 rules, which points directly at rule compaction as future work.

7.6 How Rule-Based Output Supports Interpretability

The practical difference between the LCS and the conventional models is what can be said about an individual prediction. Random Forest (B) outperformed the improved eLCS, but it can only offer aggregate feature importances; it cannot state why a particular respondent was flagged. With the LCS, any prediction can be traced to the specific rules that matched that record, each readable as a statement about a subgroup.

This advantage should not be overstated. The population contains 1,821 rules, so the model as a whole is not simple even though each rule is. The LCS provides local interpretability, an account of individual decisions, rather than global simplicity.

The limits also matter in practice. The improved system detects 76.2% of cases at 29.1% precision, so roughly two in three flagged people do not have diabetes. Combined with the uneven rule quality, this suggests the system is suited to prompting a confirmatory blood test, and to helping a clinician see which characteristics drove that prompt. It is not suitable for diagnosis, and it should not be used for any consequential decision such as insurance pricing or employment screening.

Task 8: Discussion, Limitations, and Future Work

8.1 Reasons for Performance Improvement

The main improvement in eLCS performance resulted from preprocessing and balancing the training data. The original eLCS model trained on the minimally processed, imbalanced dataset achieved an accuracy of 0.8606 but a balanced accuracy of only 0.5000, with zero recall for the positive class. Its apparently high accuracy was caused by predicting almost every participant as having no diabetes.

When the original eLCS was trained using the balanced preprocessed data, balanced accuracy increased to 0.7302. The improved eLCS achieved a slightly higher balanced accuracy of 0.7309 after attribute configuration and validation-based hyperparameter selection. This additional increase of approximately 0.0008 was small, and the difference between the original and improved eLCS models on the preprocessed data was not statistically significant after Holm correction. Consequently, most of the performance improvement can be attributed to class balancing and preprocessing rather than modification of the eLCS configuration.

8.2 Practical Meaning of the Results

The improved eLCS achieved a recall of 0.7621, meaning that it identified approximately 76% of participants in the positive diabetes risk class. This represents a substantial practical improvement over the original baseline, which identified no positive cases. Although overall accuracy decreased to 0.7085 and precision was only 0.2913, this trade-off may be acceptable for preliminary screening, where failing to identify an at-risk participant may be more harmful than recommending further assessment for someone who is not ultimately diagnosed.

The improved eLCS did not outperform the strongest conventional models. Random Forest B achieved a balanced accuracy of 0.7486, compared with 0.7309 for the improved eLCS. However, eLCS provides human readable rules that can help explain which combinations of characteristics influenced its predictions. This interpretability may be valuable when predictions need to be reviewed, although the rules should support, not replace, clinical judgement. The system should therefore be considered a risk-screening aid rather than a diagnostic tool.

8.3 Limitations

The dataset is based largely on self-reported health and lifestyle information, making it vulnerable to recall error, response bias and inaccurate reporting (Althubaiti, 2016). Although the dataset is large, its records originate from a particular survey population and period, so the results may not generalise to other countries, healthcare settings or more recent populations. The available variables are also limited to the collected survey features and do not include clinical measurements such as blood glucose, HbA1c or detailed medical history.

Class imbalance remained an important limitation. Random undersampling produced a balanced training set but discarded many majority class records, potentially removing useful information about participants without diabetes. SMOTENC preserved more information as a secondary comparison, but synthetic records may contain uncommon or clinically implausible combinations. The unchanged test set remained imbalanced, which explains why improvements in recall were accompanied by relatively low precision and many false-positive predictions.

Data cleaning reduced missingness and inconsistent values, but it cannot remove all measurement noise or correct inaccurate original responses. Furthermore, associations learned from the selected features do not establish causal relationships. Some eLCS rules may therefore reflect correlations specific to this dataset rather than reliable clinical knowledge.

8.4 Future Work

Future work should evaluate alternative imbalance handling methods that retain more majority class information while avoiding unrealistic synthetic records. This could include repeated undersampling, ensemble-based balancing or cost sensitive eLCS variants. Broader hyperparameter optimisation and alternative LCS variants could also be investigated to determine whether rule quality and predictive performance can be improved beyond the limited configurations evaluated in this project.

The system should be tested using independent and more recent datasets to assess whether its performance generalises beyond the current survey population. Additional clinically relevant features, such as blood glucose measurements, medication history and family history, may improve risk estimation if suitable data become available. Performance should also be assessed across demographic groups to identify potential disparities.

Finally, the learned eLCS rules should be reviewed by healthcare or domain experts to determine whether they are clinically meaningful, understandable and safe to use. Any future deployment should include ongoing monitoring and clear communication that the system provides screening support rather than a medical diagnosis.

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Appendices

Appendix A: GitHub Repository Link:

https://github.com/ikhalf0/Data_Engineering_and_Ai

Appendix B: Link for Commit History

https://github.com/ikhalf0/Data_Engineering_and_Ai/commits/main